

Artificial Intelligence

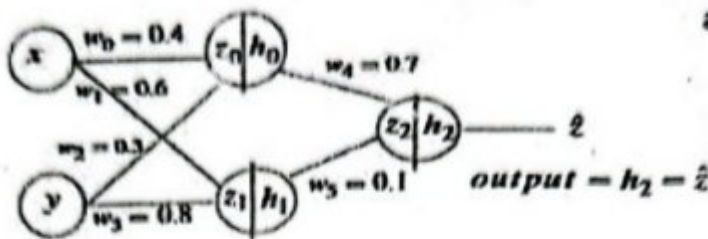
Question Bank — CT-mid-2-alternative

7th — Question 5

5. (a) Define neuron. Describe the mathematical model of a neuron. [4]
- (b) Discuss the classification approaches to pattern recognition. [4]
- (c) Discuss about various types of knowledge to build a neural network. [3]
- (d) What do you mean by learning? Write short note about reinforcement learning. [4]

5th — Question 5

5. (a) Define forward and backward reasoning/chaining. What are the factors that determine whether it is better to reason forward or backward? [4]
- (b) Define neuron. Describe the mathematical model of a neuron. [4]
- (c) Review the following neural network and compute Z: [4]



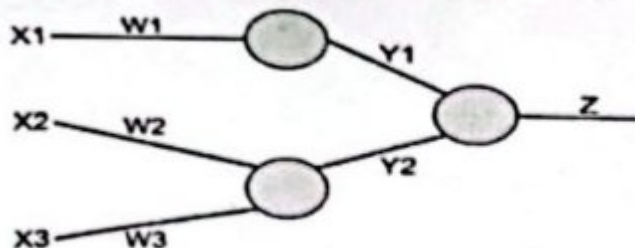
Where $x=1$, $y=1$ and other values are given,

- i) Compute the value of Z without a transfer function.
- ii) Compute the value of Z with the Mean Squared Error function E , used at all the neurons.

- (d) What do you mean by learning? Write short note about reinforcement learning. [3]

4th — Question 5

1. (a) Write the difference between Universal Quantifiers and Existential Quantifier with example.
- (b) What is knowledge representation? How can you represent knowledge base using propositional logic?
- (c) Define neuron. Describe the mathematical model of a neuron.
- (d) Review the following neural network and compute Z:

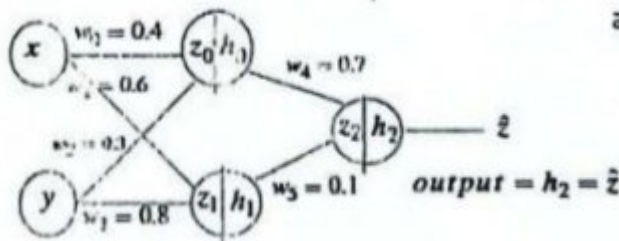


Where $X_1=17$, $X_2=7$, $X_3=12$, $W_1=0.6$, $W_2=0.3$, $W_3=0.1$, weight for $Y_1=0.6$, Weight for $Y_2=0.45$

- Compute the value of Z without a transfer function.
- Compare the value of Z with a threshold function. If the value is 5 or less, call it 0, otherwise call it 1.
- Compute the value of Z with the sigmoid transfer function used at all the neurons.

7th — Question 6

- Define forward and backward reasoning/chaining. What are the factors that determine whether it is better to reason forward or backward?
- Define neuron. Describe the mathematical model of artificial neural network.
- Review the following neural network and compute Z :

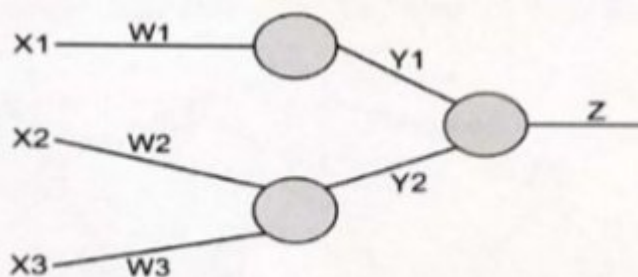


Where $x=1$, $y=1$ and other values are given,

- Compute the value of Z without a transfer function.
- Compute the value of Z with the Mean Square Error Function E , used at all the neurons.
- Explain Syntactic processing and Semantic analysis in Natural Language Processing.

6th — Question 6

- What do mean by neuron of human brain. Describe the mathematical model of a neuron.
 - Review the following neural network and compute Z :



Where $X_1=17$, $X_2=7$, $X_3=12$, $W_1=0.6$, $W_2=0.3$, $W_3=0.1$, weight for $Y_1=0.6$, Weight for $Y_2=0.45$

- Compute the value of Z without a transfer function.
- Compare the value of Z with a threshold function. If the value is 5 or less, call it 0, otherwise, call it 1.
- Compute the value of Z with the sigmoid transfer function used at all the neurons.

(c) What do you mean by learning? Describe about reinforcement learning, supervised learning and unsupervised learning with proper example.

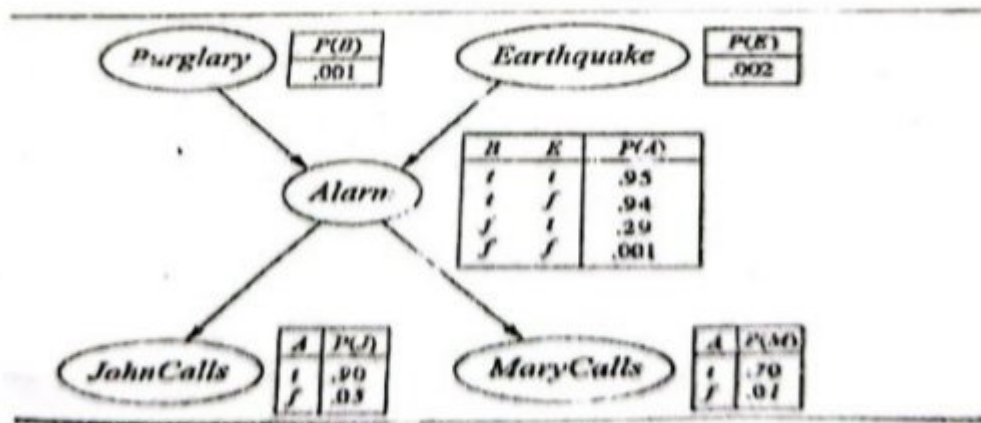
05

06

04

5th — Question 6

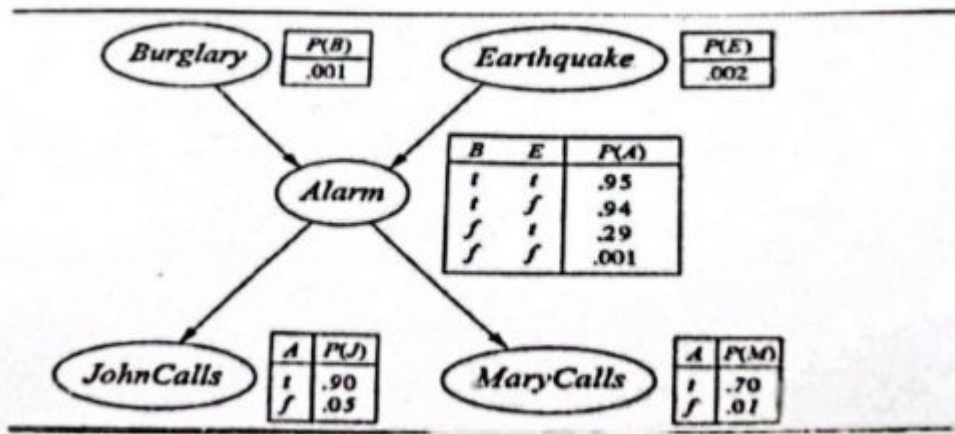
1. (a) Compare among propositional, first-order, and fuzzy logic.
- (b) Distinguish between supervised learning and unsupervised learning?
- (c) What is brain? It is faster than fastest digital computer. Explain with example.
- (d) A Bayesian network (Fig. 1) showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J and M stand for Burglary, Earthquake, Alarm, JohnCalls, and Marry Calls, respectively. The independent probability helps us to write in a simplified way the joint distribution $P(B, E, J, M)$.



- i) If no evidence is observed, are Burglary and Earthquake independent? Justify your answer by calculating whether the probabilities involved satisfy the definition of conditional independence.
- ii) Determine the probability of the event that the Alarm has sounded but neither a Fire nor an Earthquake has occurred and both John and Marry call.
- iii) Determine the probability of the event that the Alarm has sounded and Fire has occurred, an Earthquake has not occurred and both John and Marry call.

4th — Question 6

- (a) Compare among propositional, first-order, and fuzzy logic.
- (b) Do you think a planning agent can overcome the limitations of a general problem solving agent? Justify your answer.
- (c) Define forward and backward reasoning/chaining. What are the factors that determine whether it is better to reason forward or backward?
- (d) A Bayesian network (Fig. 1) showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters E, A, Y, and S stand for Burglary, Earthquake, Alarm, JohnCalls, and Marry Calls, respectively. The independent probability helps us to write in a simplified way the joint distribution $P(E, A, J, M)$.



- i) If no evidence is observed, then there is no need to perform any calculations. This is because the probabilities of independence are high.
- ii) Determine the probability of the event that the alarm has sounded but neither a fire nor an earthquake has occurred and both John and Mary call.
- iii) Determine the probability of the event that the alarm has sounded and Fire has occurred, an earthquake has not occurred and both John and Mary call.